Sefik EGRIBOZ (M.Sc., PMP, PSPO, ITIL)



TECHNICAL DELIVERY & TRANSFORMATION LEAD

Solutions Architect | Programme Delivery | Enterprise Integration

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Immediately available | Open to relocation | sefikegriboz.net

PROFESSIONAL SUMMARY

Key results: **9 systems delivered to production** | monthly incidents down from **8 to 1** on **six mission-critical systems** | new-integration lead time from **one week to one day** | **EUR 8M** cumulative savings | **98%** fewer reported ATC confusion events

Technical delivery and digital transformation lead, experienced in airline IT, with **50+ projects** managed up to a **EUR 3.5M** programme in operations that cannot stop: from mainframe modernization (as an engineer) to an **enterprise integration transformation** onto governed, event-driven, API-first platforms carrying **~50M annual transactions**.

From heuristics to production AI in a safety-critical process: callsign safety for an **NP-hard** Quadratic Assignment Problem, plus a real-time Early Warning System with zero restricted-airspace dispatches.

Transformation with **delivery accountability**: roadmaps, OKRs and KPIs for six mission-critical systems (EUR 1M budget), critical path reviews and operational handover. Customer-side product owner for NetLine, SchedConnect and flight planning across Turkish Airlines, AJet and SunExpress (a Lufthansa joint venture): a multi-vendor RFP and **54 partner carriers** in schedule exchange. Rated **exceptional** in DDI Manager Ready.

RECENT PROFESSIONAL DEVELOPMENT

Career Break and Professional Development | **Istanbul** | **Sep 2024 – Present**

- Planned break for family and structured upskilling: **Harvard's CS50 AI** (12 Python projects: search, optimization, machine learning, neural networks, NLP), AWS Certified AI Practitioner and Cloud Practitioner, PSPO I and continued research on graph-based AI for aviation safety.

PROFESSIONAL EXPERIENCE

Solutions Architect and Delivery Lead | **Turkish Technology** | **Istanbul, Turkiye** | **Jan 2023 – Sep 2024**

Official title: Distinguished Expert, the highest technical grade at Turkish Technology.

- Cut reported ATC confusion events by **98%**, from **50 to 1 per season** (sustained since), on up to ~1,500 daily peak-season flights, replacing the earlier heuristic with a fully AI-based callsign safety system aligned with EUROCONTROL CSST.
- Reframed callsign assignment as an **NP-hard** Quadratic Assignment Problem and built a machine learning solution on **Graph Neural Networks (GNN)**: **Graph Attention Networks (GAT)** on Neo4j with PyTorch Geometric and feature engineering.
- Owned the transformation roadmap and technical direction for the operational platform estate, with transformation governance that tied roadmap, architecture and delivery decisions to **integration dependency management**.
- Reduced monthly incidents from **8 to 1** in 10 months on **six mission-critical operational systems** (1,000 users, ~100 concurrent, 5M annual transactions, **EUR 1M** budget) by defining their platform roadmaps, OKRs and KPIs.
- Ran the NetLine/Ops replacement selection (**5 vendors, 3 shortlisted, 2 POCs**; a **EUR 1.5M** decision) and vendor evaluation of flight planning platforms (Jeppesen, CAE formerly FPM, Lido) through RFP/RFI, vendor management and business cases.
- Advised VP-level leadership on build-versus-buy, technology strategy and long-term R&D strategy.
- Held schedule mismatches at **zero** across two carriers and two reservations systems during AJet's migration from TROYA, the group's reservations and inventory platform, to a separate third-party PSS, through a transition architecture.
- Ran SchedConnect-based synchronization, inter-carrier schedule distribution and schedule change management.
- Led an **8-person cross-functional team** (project manager, 2 system analysts, 1 business analyst, 4 software engineers), with Scrum Master support as required, and rated team members' performance.
- Delivered **9 systems to production** as architecture and design decision owner.
- Defined the implementation approach, mentored engineers hands-on and cleared complex technical blockers.
- Established design authority and design reviews bringing architecture decisions under ISO 27001 compliance, software quality management and enterprise architecture governance, and supported regulatory audits and ICAO compliance.
- Shaped integration and technology strategy for next-generation aviation operations platforms as **NetLine Subject Matter Expert** for Turkish Airlines, SunExpress and AJet, and as SME in callsign safety, IOCC and operational control systems.

Senior Integration Architect | **Turkish Technology** | **Istanbul, Turkiye** | **Jan 2021 – Jan 2023**

Official title: Senior Expert, Integration Architect.

- Drove the enterprise integration transformation: standardized **200+ integration points across 86 systems** (**~50M annual transactions**) under one integration governance and data governance model.
- Set enterprise messaging and data streaming reference patterns, adopted group-wide for interoperability, maintainability, observability and operational reliability.
- Shortened new-integration lead time from **one week to one day**, cut integration-related incidents by **99%** and retired **47 legacy point-to-point interfaces** by reusing the group-wide reference patterns.
- Achieved **99.8% availability** (planned maintenance included) and zero reported unplanned downtime on mission-critical platforms serving **500+ concurrent users**, under high availability, disaster recovery design and SLA/SLO management.
- Led cross-functional delivery management through production deployment, coordinating technical dependencies and production readiness across teams and giving technical input to steering committee meetings.
- Ran **Scrum**-based agile delivery in **Jira** and **Confluence**: user stories with acceptance criteria, backlog management, sprint planning and Definition of Ready/Done.
- Enabled **1.5M** disrupted passengers a year to re-accommodate online, even those en route to the aircraft, by extending PSS/DCS integration with the in-house PSS TROYA to irregular operations (IROPS/IRROPS) and disruption management.
- Propagated IOCC decisions to inventory, reservations and departure control as they happened.
- Eliminated dispatches into restricted airspace (**18** in the 12 months before go-live, **zero** since) by engineering a real-time Early Warning System across NetLine and FPM.
- Covered up to **~1,500 daily** peak-season dispatch legs with **72** automated checks and alarms on planned routes.
- Modernized the integration landscape onto an event-driven application and data integration platform on Apache Kafka (topic design, schema registry), IBM MQ and REST/SOAP APIs, sustaining **~50M annual transactions**.
- Applied microservices, enterprise integration patterns, API-first design and API lifecycle management, including naming conventions, API versioning and backward compatibility.
- Engineered the reliability semantics behind the estate's **99.8%** availability: publish/subscribe and asynchronous processing, idempotency in consumers, retries and exponential backoff, timeouts and error recovery.
- Built in dead-letter handling and poison-message processing, message ordering and durability, and replay of persisted streams for recovery.
- Owned the solution architecture artefacts that kept decisions traceable to a maintained target state architecture: solution blueprints, architecture roadmaps, High-Level and Low-Level Design (HLD/LLD), Architecture Decision Records (ADR) and non-functional requirements (NFR).
- Raised release frequency from **1 to 4 a month** and **halved failed releases** by setting the estate's engineering practice: CI/CD architecture and DevOps release automation.
- Gated releases with GitOps configuration control, automated testing strategies, and performance and security testing.
- Caught **99%** of integration failures before users reported them and reduced mean time to detect from **one hour to one minute** with a real-time observability tool built across all live enterprise integrations.
- Based the tool on structured logging, correlation IDs and distributed tracing, and directed delivery of Grafana dashboards, metrics and alerting to support proactive issue detection.

Project Manager, Flight Operations | **Turkish Airlines** | **Istanbul, Turkiye** | **Jun 2014 – Jan 2021**

Functional title: Lead Solutions Architect and Platform Owner.

- Designed and built **TKAODS** (Airline Operational DataStore) and evolved **OFPDS** (Operational Flight Plan DataStore, live since 2010), which together became the **single source of truth** for schedule and flight data.
- Took schedule and flight-change delivery from **next-day to near-real-time** for **78 consuming systems**, replacing **70 separate flight-data interfaces** by routing every change through both platforms, 24/7.
- Architected both as **Tier-0** integration layers in front of the Lufthansa Systems NetLine suite (NetLine/Sched, NetLine/Slot, NetLine/OPS & Base) and flight planning.
- Owned delivery scope, implementation planning, vendor activities, user acceptance testing (UAT), risk tracking and operational handover for multi-system releases across scheduling, flight planning, IOCC and enterprise systems.
- Drove technical direction, integration and platform modernization for multimillion-euro airline operational systems (**EUR 6.5M** total budget: scheduling, slot management, flight planning, IOCC) as lead architect with delivery accountability.
- Worked end to end: system and business analysis, software architecture and hands-on coding through release, operational handover and 24/7 production under ITIL.
- Owned enterprise-wide flight plan and flight data distribution for planning, operational and post-flight periods.
- Served as the single integration point for **eleven main downstream domains**: scheduling, crew, load control, PSS/DCS interfaces, cargo, maintenance, DWH, revenue and cost accounting, SAP finance, HR/payroll and regulatory reporting.
- Processed **~100K update records a day** (peak-season average) for up to **~1,500 flights a day**, 24/7 and near-real-time, across summer and winter seasons.

- Consolidated, transformed and enriched every update through source-to-target mappings, ETL/ELT pipelines and integration rules under data quality and metadata controls.
- Designed batch, near-real-time and real-time distribution patterns that absorbed **~22.5M** flight-data changes a year (~450K annual flights, ~50 changes per flight) before fan-out to large-scale distributed data processing and analytics.
- Set consumer-specific horizons, from a **rolling three-day window** to a **90-day window**.
- Fed operational changes (delays, cancellations) to passenger-notification systems, triggering **75M notifications a year**, and flight data to loyalty and disruption hotel systems, so all three ran on the same data as operations.
- Designed the enterprise operational data and integration architecture on the AODB (Airline Operational DataStore) pattern, standardizing flight, leg, schedule and operational flight-plan data airline-wide.
- Defined the logical data models, entities and relationships and canonical data structures, and applied the ESB pattern twice.
- Evolved both Tier-0 platforms for **ten years** (2014 – Sep 2024) as de facto product owner with final authority over feature prioritization and architectural evolution.
- Coordinated stakeholders and cross-team dependencies for two Tier-0 platforms and ran critical path analyses reviewed with management, carrying releases through production under ITIL-aligned **Site Reliability Engineering (SRE)** practices and continuous monitoring.
- Managed NetLine's master and reference data (NetLine/Base) with the team, keeping it owned at its source, and trained key users in master data management (MDM) to support user adoption and organisational change.
- Built a business intelligence reporting layer into both platforms on the same enriched data and fed it back into NetLine and the flight planning system.
- Gave enterprise reporting and operations one shared version of the data.
- Kept **850 NetLine users** working through the pandemic as NetLine product owner, coordinating the VMware solution.
- Received Turkish Airlines **achievement certificates** in 2017 for contributions to the Fuel Management Information System, the FPM & FE implementation and the OpsEye Diversion Manager module and its rollout.

Acting Team Leader | Turkish Airlines | Istanbul, Turkiye | Jan 2007 – Jun 2014

Functional title: Integration Solution Design Lead.

- Delivered **EUR 8M** in cumulative savings from 2012 to 2023 (**EUR 800K** in the first year) with a multi-currency overflight cost engine built on a data pipeline of country entry/exit timestamps and segment distances.
- Led a **6-person team** through the lifecycle of **100+ integrations** (requirements and system analysis, design, development, testing, documentation), harmonizing project plans and tracking cross-functional milestones per project.
- Rated team members' performance and interviewed technical candidates as Acting Team Leader; hiring decided by others.
- Applied PMP practices to scope definition, executive (C-level) stakeholder management, customer relationship management, vendor management, risk tracking, implementation planning and operational handover.
- Headed solution design and delivery for flight operations, crew and flight-planning integrations across global aviation vendors, building multithreaded real-time integration frameworks and contributing to an in-house Enterprise Service Bus (ESB) on Service-Oriented Architecture (SOA).
- Introduced SchedConnect in 2007 for inter-carrier schedule enrichment and distribution, establishing recurring schedule data exchange with **54 partner carriers** under IATA data standards.
- Architected the schedule-to-PSS (TROYA) message integration under IATA SSIM Chapter 4 (SSM) and Chapter 5 (ASM) over Type B messaging.
- Kept inventory and reservations in step with cancellations, re-timings, equipment and routing changes, reinstated flights and disruption reason codes.
- Developed the first heuristic Callsign Similarity & Assignment System (phonetic, visual and semantic similarity), the groundwork later rebuilt as the graph-based AI solution once reclassified as NP-hard.
- Commended in writing in 2010 by Turkish Airlines' **Chief Pilot and Head of Flight Operations** for flight operations projects, the callsign project first among them, and personally by the **General Manager and CEO**.

Software and Integration Engineer | Turkish Airlines | Istanbul, Turkiye | Apr 2000 – Jan 2007

Official title: Analyst Programmer.

- Drove the technical work of the first major legacy modernization from day one as an implementation team member.
- Moved schedule and operations control off the IBM mainframe (ADABAS/Natural) onto Oracle on Unix (AIX) and NetLine.
- Cut costs by **EUR 100K** a year by building the timetable publication system (Java, C, PL/SQL).
- Turned airline schedules and partner codeshare flights into a PDF timetable for web and print, so scheduling produced the timetable books itself.
- Contributed to the first production implementation of Lufthansa Systems NetLine (NetLine/Sched, NetLine/Slot, NetLine/OPS & Base), the platform that scaled with fleet growth from **60 to 500+ aircraft** and from 10M to 85M passengers.

CORE COMPETENCIES & TECHNICAL SKILLS (hands-on)

Transformation and Programme Leadership: Digital and technology transformation, transformation governance, programme management, delivery governance, legacy modernization, roadmap definition, executive (C-level) stakeholder management, customer consulting and workshop facilitation.

Change, Adoption and Benefits: Stakeholder alignment, user readiness through UAT and key-user training, operational handover, multi-carrier platform adoption; benefits realization through delivered and measured outcomes (EUR 8M cumulative savings, 98% fewer reported ATC confusion events, 99.8% availability).

Technical Platforms, Data & Architecture: Enterprise Data Architecture, Data Modeling, Canonical Data Model, Operational Data Store (ODS), Data Integration Architecture, Data Pipeline Architecture, Event-Driven Architecture, API Architecture, Apache Kafka, IBM MQ, Enterprise Service Bus (ESB), REST and SOAP APIs.

Aviation Operations and Airline IT: Integrated Operations Control Center (IOCC), operational and tactical flight planning (Jeppesen; FPM, now CAE Flight Planning Solutions; Lido), schedule management; DWH, SAP and MRP interfaces.

IATA Schedule Data Standards: SSIM Chapter 7 (Presentation and Transfer of a Schedule Data Set); schedule-to-PSS distribution across three group carriers.

Analytics & Graph / Advanced Analytics and AI: Neo4j and graph databases, PyTorch and PyTorch Geometric, MLOps, data quality and metadata management.

Engineering Platforms & Languages: Oracle DB, Docker and containerization, Linux (RHEL) and Unix administration (AIX, HP-UX), application administration on Windows Server; Languages: Python, Java, C, PL/SQL, shell scripting (Bash, KSH), YAML and JSON; Git and Jenkins; UML; multithreaded real-time and database-driven systems; incident analysis and performance tuning.

EDUCATION

• Ph.D. (coursework)	Business Administration (Finance)	Yeditepe University	Istanbul
• M.Sc.	Management Engineering; thesis on data warehousing for mass customization	Istanbul Technical University	Istanbul
• B.Sc.	Computer Science Engineering	Istanbul University	Istanbul

LANGUAGES

• English	C1	Full Professional
• German	A2	Studying towards B1
• Turkish		Native

CERTIFICATIONS

• Project Management Professional (PMP), Credential ID 1225730	PMI	2008
• Professional Scrum Product Owner I (PSPO I), Credential ID 1347372	Scrum.org	2026
• ITIL v3 Foundation, Candidate No. 990453483	BCS, The Chartered Institute for IT	2013
• AWS Certified Cloud Practitioner (CLF-C02)	Amazon Web Services (AWS)	2026
• AWS Certified AI Practitioner (AIF-C01)	Amazon Web Services (AWS)	2026
• CS50's Introduction to Artificial Intelligence with Python (12 projects)	Harvard University (CS50)	2026
• Manager Ready (DDI) leadership simulation: exceptional rating	TAG Taurus Group	2024
• AI, Cloud and Container Foundations: OCI AI Foundations; Agentic AI Foundations; Microsoft Azure AI Essentials; IBM SkillsBuild; PMI AI (CPMAI, GenAI); Docker Foundations Professional Certificate	Oracle, Microsoft, IBM, PMI, Docker	2026
• Lufthansa Systems SchedConnect vendor training (two weeks)	Lufthansa Systems	2007